TAO Wenwen

Curriculum Vitae — PhD and Graduate Admissions

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Research Profile

Clinically trained multimodal human neuroscience researcher studying youth mental health and developmental variation in brain–behavior relationships. Patient-facing work first taught me how clinical heterogeneity complicates measurement; addiction cohorts then taught me how recruitment, follow-up, acquisition, quality control and early preprocessing determine whether a scientific question can be answered.

Subsequent depression neuroimaging training deepened my analysis and interpretation skills. I now work from the scientific question backward—independently designing synchronized EEG–fNIRS, physiological, behavioral and clinical research systems around downstream modeling needs. I am particularly interested in distinguishing stable vulnerability from within-person state change and using longitudinal multimodal data to inform individualized assessment and intervention.

Research Interests

Youth mental health and developmental psychopathology · Multimodal brain–behavior dynamics · Within-person and longitudinal variation · Emotion regulation, cognitive control, risk and resilience

Education

B.M. in Clinical Medicine | Nanchang University **2017–2022**

GPA 3.2/4.0 (82/100) · Psychiatry, neurology, medical statistics and evidence-based medicine

Research Experience

Research Assistant | Tsinghua University, CogDrive Lab **Mar 2026–Present**

State Key Laboratory of Intelligent Green Vehicle and Mobility · Advisor: Prof. Dongpu Cao · Clinical collaboration: Tongji Hospital, HUST

Multimodal developmental neuroscience of youth depression and translation to intelligent health-cockpit research

- Scientific question: How can neural, hemodynamic, autonomic, behavioral and clinical signals characterize mental and emotional states across ages 6–18 while distinguishing developmental differences from clinically meaningful variation?
- Research ownership: Independently translated this question into the study protocol, cognitive-affective paradigms and clinician-led interview workflow for a planned 300-participant clinical cohort and 150 healthy controls; the protocol received joint scientific review and hospital IRB approval.
- Methodological solution: Designed synchronized acquisition and event-trigger architecture integrating simultaneous 64-channel EEG –fNIRS, eye tracking, facial video, speech, ECG, PPG, GSR and respiration during computerized tasks and clinical interviews, with acquisition planned around downstream preprocessing, neurovascular-coupling and multimodal modeling requirements.
- Established age-adaptive clinical ground truth using CDRS-R, self/parent reports, K-SADS-PL DSM-5 confirmation and multisource severity grading, including developmentally appropriate consent, assent and risk-escalation procedures.
- Built BPlus MVP-0 to operationalize Rest, MMN, VFT, N-back, Emotion and Recovery modules with response/event logging, preflight checks and QC/state reports. Software demonstrations and dry runs passed; real-hardware synchronization and formal pilot data collection remain pending.
- Designed the health-cockpit translation pathway to connect validated laboratory markers with lower-burden in-vehicle sensing, confidence-aware outputs and safety-gated human–AI interaction; clinical validation and vehicle deployment remain future stages.

Research Intern | Tsinghua University, Yan Lab **Sep 2025–Mar 2026**

Advisor: Prof. Chaogan Yan · Mind Flower depression research

Depression neuroimaging, repeated symptom measurement and individualized rTMS

- Scientific question: Examined how individual differences in brain function, cognition and repeated symptom trajectories could inform understanding of depression and treatment response, combining clinical phenotyping, MRI, cognitive tasks, rTMS records and wearable follow-up.
- Participant and acquisition role: Conducted MINI and serial HAMD/HAMA/MADRS assessments and managed resting-state, rumination and distraction MRI sessions, including individualized cue preparation, E-Prime task adaptation, head-motion monitoring (<3 mm) and OptiMRI audio QC.
- Analytical development: Completed structured hands-on training in DPABI/DPABISurf and DeepHCP neuroimaging workflows and applied this training to resting-state and task-fMRI preprocessing, quality control and participant-level interpretation, moving beyond the basic QC work of earlier projects.
- Linked repeated HAMD-6 and rTMS tolerability records with WCST, SST, emotional n-back and two-week HRV, sleep and activity monitoring; standardized E-Prime recovery, export and hierarchical NAS archiving across repeated visits.

Visiting Research Intern | Beijing Normal University, Liu Lab **Sep 2025–Mar 2026**

Advisor: Dr. Zheyun Liu

MRI-guided transcranial ultrasound stimulation of bilateral GPi in Parkinson's disease

- Supported individualized bilateral GPi targeting using structural MRI and Brainsight neuronavigation within a multidisciplinary TUS workflow.
- Operated an approximately 1.5-hour TUS–cognitive assessment protocol focused on memory and cognitive flexibility, documenting pulse parameters, behavioral responses, safety and participant-reported auditory/thermal tolerability.

Research Assistant | Shenzhen Mental Health Center, Addiction Medicine **Jan 2024–Aug 2025**

Advisors: Prof. Mei Yang, Prof. Xiaojian Jia, Dr. Gengdi Huang and Dr. Yi Zhang

Addiction neuroimaging and longitudinal clinical cohorts

- Scientific question: Investigated how clinical, circuit-level and biological factors relate to addiction severity and longitudinal outcomes in alcohol and methamphetamine use disorders.
- Clinical-cohort responsibility: Coordinated recruitment, screening, psychiatric assessment and longitudinal follow-up for 290+ participants. Worked with clinicians and department leadership to identify eligible participants, and with radiology staff and technical engineers to prepare MRI/DTI, EEG, behavioral and biospecimen workflows; maintained >95% data completeness.
- Early technical development: Participated in static and task-based MRI/DTI and EEG acquisition, quality control and basic preprocessing, while building participant, sample, metadata and anomaly-tracking procedures that made downstream analysis reproducible.
- Research progression: This work established my foundation in running real clinical cohorts and showed me that acquisition decisions, participant communication, follow-up and data provenance are inseparable from later modeling and interpretation.

Selected Research Projects and Contributions

Gut Microbiota–Metabolite–Circuit Mechanisms in AUD | 220 participants | *Can microbiome and metabolic disruption be linked to white-matter injury and craving?*

- Integrated 16S rRNA microbiome profiles, GC–MS short-chain fatty-acid measures and DTI/fMRI to test a cross-scale gut–brain pathway rather than treating each modality as a separate endpoint.
- Identified three candidate microbial markers associated with SCFA levels and observed correlations between lower SCFAs and lower FA in the anterior limb of the internal capsule and corpus callosum; these associations require longitudinal and mechanistic confirmation.

Neuroprotective Mechanisms of TMP in AUD | 160 participants | *Does TMP intervention relate to change in white-matter microstructure?*

- Coordinated multimodal imaging and compared pre/post FA in the corpus callosum and superior longitudinal fasciculus; preliminary differences require longitudinal confirmation.
- Optimized the combined DTI + resting-state fMRI acquisition and quality-control workflow, including susceptibility-artifact reduction in basal-ganglia imaging.

MA-Related Frontotemporal Injury and Risky Decision-Making | 10 newly acquired cases + 80-case database | *How do structural and task-state signals relate to cognitive impairment and risky choice in methamphetamine dependence?*

- Independently acquired MRI/EEG data from 10 participants and maintained the broader clinical database; contributed to DTI/fMRI analyses of prefrontal–hippocampal, cingulum and fornix pathways.
- Led participant-held-out prediction analyses comparing task state with pre-decision EEG; the resulting first-author manuscript found dominant task-state effects and limited incremental predictive value from pre-decision EEG.

PFC–Ventral Striatal Executive-Control Network | 20 deeply phenotyped cases | *Can structural–functional circuit measures help explain executive-control deficits and relapse risk?*

- Examined PFC–ventral striatal structural connectivity with DTI and task-fMRI functional coupling, then developed an exploratory logistic-regression relapse model.
- Used the findings to generate testable hypotheses for future connectivity-informed neuromodulation rather than presenting the exploratory model as a validated clinical predictor.

Publications and Manuscripts

- Tao, W. Early Depressive Symptom Severity and Six-Month Return to Drinking After Inpatient Treatment for Alcohol Dependence: A Prospective Cohort Study. Manuscript submitted to Psychiatry Research. First author (not co-first author); prospective cohort and longitudinal outcome analysis.
- Tao, W. Temporal Dataset Shift Limits Relapse-Risk Stratification Across the Inpatient-to-Community Transition in Alcohol Dependence: A Temporal Comparison of Conventional Machine Learning and Self-Supervised Representation Learning. Submitted to Addiction Science & Clinical Practice (ASCP). First author (not co-first author); temporal validation and comparative modeling.
- Tao, W. Task state dominates participant-held-out prediction of risky decisions in methamphetamine dependence, with limited incremental value from pre-decision EEG. Submitted to Addictive Behaviors. First author (not co-first author); participant-held-out EEG and behavioral prediction analysis.

Research Direction

- Immediate empirical focus: Use longitudinal, clinically grounded multimodal studies to distinguish stable vulnerability from within-person state change in youth mental health and development.
- Mechanistic focus: Examine how control, reward, memory, affective and physiological systems dynamically coordinate across context, symptom state and developmental stage, including neurovascular coupling measured with simultaneous EEG–fNIRS.
- Long-term translation: Develop interpretable, state-sensitive assessment and intervention frameworks; treat BCI, closed-loop neurotechnology and human–AI systems as later extensions of the same state-to-action question rather than separate research identities.

Clinical Experience

Resident Physician | Wannian County Hospital of TCM **Dec 2022–Dec 2023**

- Managed 15–20 chronic-disease inpatients through assessment, treatment planning and follow-up, developing first-hand understanding of heterogeneity in real-world clinical presentations and the need for clinically interpretable research measures.

Clinical Intern | Ganzhou People's Hospital **Feb 2021–Feb 2022**

- Completed 12 rotations, including Neurology ICU and Cardiology ICU; performed supervised lumbar puncture and ECG interpretation and participated in routine inpatient and perioperative workflows.

Methods and Technical Expertise

Multimodal experimental design: Protocol and paradigm development; TTL/LSL and event-trigger architecture; synchronized acquisition; clinical interview workflow; multimodal QC; analysis-ready data specification

Neuroimaging analysis: Structural MRI; resting-state and task fMRI; DTI tractography; FA/ADC; structural–functional connectivity; DeepHCP; DPABI/DPABISurf; FSL; MRI-based targeting

Neurophysiology and physiology: EEG (32/64-channel); simultaneous EEG–fNIRS; ECG; PPG; GSR; respiration; eye tracking; signal QC; exploratory neurovascular coupling

Clinical and longitudinal research: MINI; K-SADS-PL DSM-5; CDRS-R; HAMD/HAMA/MADRS; recruitment; repeated assessment; participant follow-up; cohort and IRB coordination; pediatric safeguards

Quantitative and computational methods: Python; MATLAB; R; SPSS; regression/classification; cross-validation; AUC; participant-held-out and temporal validation; longitudinal/repeated-measures analysis

Translational systems: BPlus MVP-0; response/event logging; preflight and QC reporting; longitudinal databases; Data Dictionary development; multicenter harmonization; health-cockpit research architecture

Additional modalities: rTMS; TUS; Brainsight neuronavigation; 16S rRNA microbiome sequencing; GC–MS SCFA metabolomics; biospecimen SOPs

English — TOEFL 97; IELTS 6.5 | Chinese — Native